

Proximal Balancing

Working Draft

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We study causal adjustment with observed pretreatment covariates X , a high-dimensional pretreatment proxy $W = (W_1, \dots, W_J)$, and an unobserved confounder. For one prespecified block j , the proposed cross-view construction learns a possibly stochastic, discrete or continuous representation $Z_j = \phi_j(V_j, \epsilon_j)$, with output dimension no larger than that of $V_j = (X, W_{-j})$, without using the held-out block W_j , and measures treatment imbalance through the conditional distribution of (X, W_j) . We prove an exact bias decomposition and show that exact cross-view balance identifies the population average treatment effect under joint latent exchangeability, held-out-view separation, overlap, and outcome-relevant completeness. When balance is approximate, a uniform outcome-relevant stability condition yields the sensitivity certificate $|\tau_{Z_j} - \tau| \leq \Gamma_j D_j(\phi_j)$. A finite-class, finite-strata learning analysis and an honest augmented inverse probability weighted estimator give an end-to-end finite-sample bound that separates representation error, hidden-confounding sensitivity, nuisance error, and evaluation-sample variation. The causal interpretation depends on the stated cross-view structural conditions; the existence of one generic correlated proxy is not sufficient by itself.

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1. Introduction

We call our method *PROxy Blockwise Exclusion* (PROBE). PROBE withholds one prespecified block of a high-dimensional pretreatment proxy, constructs an adjustment representation from the observed covariates and the remaining proxy blocks, and uses the withheld block to evaluate residual treatment imbalance. The term *exclusion* emphasizes that the evaluated proxy block is kept out of the representation input, thereby preventing literal self-copying.

1.1. Related Works

Balancing scores and covariate balancing. The propensity score is the canonical reduction of observed pretreatment covariates: under treatment ignorability given the measured covariates, conditioning on the treatment probability balances those covariates across treatment arms and identifies causal effects (Rosenbaum and Rubin, 1983). Entropy balancing and the covariate-balancing propensity score instead choose weights or treatment models that target balance directly (Hainmueller, 2012; Imai and Ratkovic, 2014). These methods motivate our optimization target, but their usual causal guarantee still requires the measured covariates to suffice for confounding control. PROBE instead allows a latent confounder and asks what a held-out proxy block can reveal about the bias remaining after reducing the other pretreatment measurements.

Causal representation learning. Representation-learning methods compress high-dimensional covariates while penalizing imbalance between treated and control representations. For example, counterfactual regression bounds counterfactual prediction error using factual prediction error and an integral probability metric between learned treatment-arm representations (Shalit et al., 2017). PROBE shares the use of a learned low-dimensional representation and a distributional balance penalty. Its distinguishing operation is blockwise exclusion: one proxy block is held out, the representation is learned from the remaining pretreatment information, and conditional treatment balance is evaluated with the held-out block.

Negative controls and proximal causal inference. Negative-control variables can reveal residual confounding because they should not be causally affected by treatment or should not causally affect the outcome, while remaining associated with hidden common causes (Lipsitch et al., 2010). Proximal causal inference turns this diagnostic idea into point identification by assigning distinct treatment-inducing and outcome-inducing proxy roles and imposing completeness and bridge conditions (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). Semiparametric proximal theory then develops influence functions and efficient estimators for the resulting identifying functionals (Cui et al., 2024). PROBE takes a different operational route: the remaining proxy blocks form the input to an adjustment representation, while the held-out block measures residual treatment imbalance. An outcome-relevant completeness condition transfers exact observed balance to the potential-outcome means needed for the population average treatment effect.

Single-proxy identification. Single Proxy Control shows that one negative-control outcome can identify the effect of treatment on the treated under a counterfactual proxy condition and suitable bridge structure (Park et al., 2024). This differs from our population average treatment effect target, our latent-confounder proxy interpretation, and our use of an observable balance discrepancy. The comparison makes clear that “one proxy” does not by itself identify a common estimand. Identification depends on what the proxy measures, which conditional independence is imposed, and whether the analysis uses a bridge or an adjustment representation checked by a

held-out proxy block.

Kernel discrepancies and sensitivity certification. Maximum mean discrepancy is an integral probability metric that equals zero only for identical distributions when its kernel is characteristic (Gretton et al., 2012). PROBE uses conditional treatment-arm discrepancies of the measured covariates and the held-out block as observable magnitudes, not kernel-test p -values. Multiplying this discrepancy by an outcome-relevant stability constant yields a bias certificate and a representation-selection guarantee. The individual ingredients have close precedents in balancing, representation learning, kernel testing, and proximal inference. We therefore state the contribution through the precise assumptions, estimand, and guarantees below rather than making an unsupported priority claim.

2. Problem Setting

We observe independent copies

$$O_i = (X_i, W_i, A_i, Y_i), \quad i = 1, \dots, n, \quad (1)$$

where $X \in \mathcal{X}$ is a possibly high-dimensional vector of observed pretreatment covariates, $\mathcal{X}$ is its value space, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome.

For each $a \in \{0, 1\}$, the *potential outcome* $Y(a)$ is the outcome that would be observed if treatment were set to a . Our target is the population average treatment effect

$$\tau \triangleq \mathbb{E}\{Y(1) - Y(0)\}. \quad (2)$$

Let U denote an unobserved pretreatment variable that may affect both treatment A and outcome Y . When U affects both A and Y , the observed-data distribution alone does not identify τ without additional structure. We therefore observe a proxy measurement W that contains information about U . The proxy is divided into prespecified blocks,

$$W = (W_1, \dots, W_J). \quad (3)$$

Each block may be scalar or vector-valued. A block can represent one repeated measurement, sensor channel, laboratory panel, or prespecified region of a larger measurement.

For each block index $j \in \{1, \dots, J\}$, we call W_j the *held-out proxy block* and let $\mathcal{W}_j$ denote its value space. We call the combination of all other blocks the *remaining proxy block* and write

$$W_{-j} \triangleq (W_1, \dots, W_{j-1}, W_{j+1}, \dots, W_J) \quad (4)$$

for the resulting vector of remaining proxy blocks. Let $\mathcal{V}_j$ denote the value space of the pair (X, W_{-j}) . Finally, define

$$V_j \triangleq (X, W_{-j}) \in \mathcal{V}_j. \quad (5)$$

The variable V_j is the combined vector of the observed covariates X and the remaining proxy blocks W_{-j} .

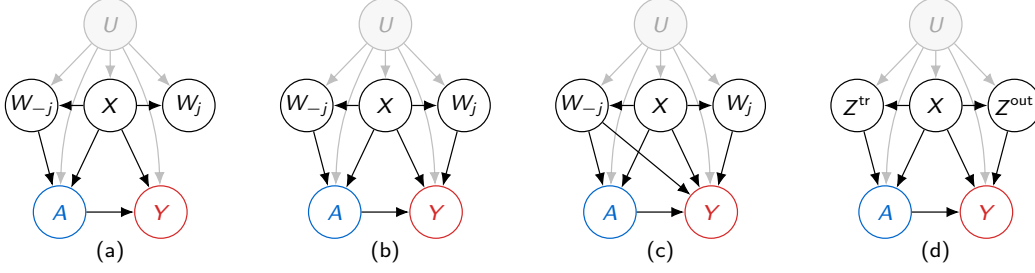


Figure 1: PROBE-compatible causal structures in panels (a)–(c) and a standard two-proxy proximal structure in panel (d).

3. Identification

Let d_{V_j} denote the coordinate dimension of V_j . Fix a user-specified output dimension $d_{Z_j} \leq d_{V_j}$. For a measurable representation function, define

$$\phi : \mathcal{V}_j \longrightarrow \mathbb{R}^{d_{Z_j}}, \quad Z_j^\phi \triangleq \phi(V_j). \quad (6)$$

Let Φ_j denote a prespecified class of such representation functions. When a representation function $\phi_j \in \Phi_j$ is fixed, we abbreviate $Z_j^{\phi_j} = \phi_j(V_j)$ as Z_j .

3.1. Identification for a Prespecified Held-Out Block

3.1.1. Assumptions

Assumption 1 (Consistency and integrability). *For each $a \in \{0, 1\}$,*

$$Y = Y(A), \quad \mathbb{E}|Y(a)| < \infty. \quad (7)$$

Assumption 2 (Latent exchangeability and a common held-out channel). *For each $a \in \{0, 1\}$,*

$$Y(a) \perp\!\!\!\perp A \mid (U, V_j), \quad (8)$$

$$W_j \perp\!\!\!\perp A \mid (U, V_j). \quad (9)$$

The first clause says that (U, V_j) is sufficient to remove hidden confounding between treatment and each potential outcome. The second clause says that, after (U, V_j) is fixed, treatment status provides no additional information about the distribution of W_j .

Figure 1 shows three PROBE-compatible examples and one representative standard proximal structure.

We then introduce the latent and proxy objects within each representation stratum.

Definition 1 (Latent and proxy objects within a representation stratum). Let P_{Z_j} denote the distribution of Z_j . For P_{Z_j} -almost every z , fix versions of the following conditional laws. Define

$$Q_{a,j,z} \triangleq \mathcal{L}(V_j, U \mid A = a, Z_j = z), \quad a \in \{0, 1\}, \quad (10)$$

and define the common conditional law of the held-out block by

$$K_j(\cdot \mid v, u) \triangleq \mathcal{L}(W_j \mid V_j = v, U = u). \quad (11)$$

For any probability law Q on (V_j, U) , let $\mathcal{K}_j Q$ denote the induced law of (X, W_j) obtained by first drawing $(V_j, U) = (X, W_{-j}, U) \sim Q$, then drawing $W_j \sim K_j(\cdot \mid V_j, U)$, and finally retaining (X, W_j) .

For each $a \in \{0, 1\}$, define

$$\mu_{a,j,z}(v, u) \triangleq \mathbb{E}\{Y(a) \mid V_j = v, U = u, Z_j = z\}. \quad (12)$$

For P_{Z_j} -almost every z , let $\mathcal{Q}_{j,z}$ be the collection of probability laws Q on (V_j, U) such that $Q \ll \mathcal{L}(V_j, U \mid Z_j = z)$ and $\mathbb{E}_Q[\mu_{a,j,z}(V_j, U)] < \infty$ for both $a \in \{0, 1\}$.

Assumption 3 (Exact outcome-relevant completeness). For P_{Z_j} -almost every z , every $Q, Q' \in \mathcal{Q}_{j,z}$, and each $a \in \{0, 1\}$,

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q'}\{\mu_{a,j,z}(V_j, U)\}. \quad (13)$$

3.1.2. Exact Identification for a Fixed Block

The next theorem gives the central identification result. It states when adjustment for Z_j identifies the population average treatment effect.

Theorem 1 (Causal effect identification using a proxy). Fix a prespecified block $j \in \{1, \dots, J\}$. Suppose Assumptions 1–3 hold for this block and its representation. If $Z_j = \phi_j(V_j)$ is constructed such that

$$0 < P(A = 1 \mid Z_j) < 1 \quad \text{almost surely}, \quad (14)$$

and

$$(X, W_j) \perp\!\!\!\perp A \mid Z_j, \quad (15)$$

then

$$\tau = \mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)]. \quad (16)$$

The theorem is a conditional mean identification result. It establishes that the conditional mean of $Y(a)$ given (A, Z_j) does not depend on A , and consistency converts this equality into the displayed adjustment formula. It does not assert the full conditional independence $Y(a) \perp\!\!\!\perp A \mid Z_j$.

3.1.3. Comparison with the Canonical Two-Proxy Proximal Model

Graphically and structurally, PROBE differs from the standard proximal causal inference framework (Miao et al., 2018; Tchetgen Tchetgen et al., 2024). The standard framework prespecifies a treatment-inducing proxy Z^{tr} and an outcome-inducing proxy Z^{out} . By contrast, PROBE requires only the held-out block W_j to satisfy $W_j \perp\!\!\!\perp A \mid (U, V_j)$. The remaining block is not subject to this restriction and may affect A , Y , both, or neither, as illustrated by panels (a)–(c).

The underlying assumptions and identification strategies are also different. A representative canonical completeness condition is

$$\mathbb{E}\{g(U) \mid Z^{\text{tr}}, A, X\} = 0 \implies g(U) = 0 \text{ almost surely} \quad (17)$$

for functions g in the specified class. Identification then proceeds by seeking an outcome bridge h satisfying an observed conditional-moment equation such as

$$\mathbb{E}(Y \mid Z^{\text{tr}}, A, X) = \mathbb{E}\{h(Z^{\text{out}}, A, X) \mid Z^{\text{tr}}, A, X\}. \quad (18)$$

PROBE instead combines observable balance in Theorem 1 with outcome-relevant completeness in Assumption 3. For comparison, the canonical proximal conditions refer to proxy validity, existence of an integrable outcome bridge, and treatment-proxy completeness for a designated proxy pair. The PROBE conditions refer to Assumptions 1–3, representation overlap, and exact balance for a designated proxy split. A natural question is whether either framework’s sufficient conditions contain those of the other.

Proposition 1 (Non-nesting of canonical proximal and PROBE sufficient conditions). There exists a causal data-generating process for which the canonical proximal conditions hold for a designated proxy pair but the PROBE conditions fail for a designated proxy split. Conversely, there exists a causal data-generating process for which the PROBE conditions hold for a designated proxy split but the canonical proximal conditions fail for a designated proxy pair.

3.2. Identification When the Valid Held-Out Block Is Unknown

Section 3.1 fixes the held-out block in advance. We now allow its composition to be unknown. Consider every nonempty proper subset

$$\mathfrak{S} \triangleq \{S \subset \{1, \dots, J\} : 1 \leq |S| \leq J - 1\}. \quad (19)$$

A subset S and its complement S^c are distinct because they exchange the held-out and remaining roles. For each $S \in \mathfrak{S}$, let $W_S = (W_j)_{j \in S}$ be the held-out vector, let $W_{-S} = (W_j)_{j \notin S}$ be the remaining vector, and define

$$V_S \triangleq (X, W_{-S}), \quad Z_S \triangleq \phi_S(V_S), \quad (20)$$

using the same construction as for a single held-out block.

For each $S \in \mathfrak{S}$, define, whenever it is well-defined,

$$\theta_S \triangleq \mathbb{E}[\mathbb{E}(Y \mid A = 1, Z_S) - \mathbb{E}(Y \mid A = 0, Z_S)]. \quad (21)$$

Theorem 1 applies to each proxy split S after replacing (V_j, W_j, Z_j) by (V_S, W_S, Z_S) throughout. Thus, if Assumption 1 and the grouped versions of Assumptions 2 and 3 hold, representation overlap holds, and $(X, W_S) \perp\!\!\!\perp A \mid Z_S$, then $\theta_S = \tau$.

The balance condition is observable at the population level. Define the population balance-screen set

$$\mathcal{B} \triangleq \{S \in \mathfrak{S} : \theta_S \text{ is well-defined in } \mathbb{R}, 0 < P(A = 1 \mid Z_S) < 1 \text{ a.s.}, (X, W_S) \perp\!\!\!\perp A \mid Z_S\}, \quad (22)$$

Each $S \in \mathcal{B}$ defines one retained proxy split and one candidate effect θ_S through Equation (21). The next result considers the case in which the true effect τ occurs more often than any other candidate value.

Theorem 2 (Identification with an unknown valid block). Suppose $\mathcal{B} \neq \emptyset$ and

$$|\{S \in \mathcal{B} : \theta_S = \tau\}| > \max_{\substack{c \in \mathbb{R}, c \neq \tau \\ \{S \in \mathcal{B} : \theta_S = c\} \neq \emptyset}} |\{S \in \mathcal{B} : \theta_S = c\}|, \quad (23)$$

This condition means that more proxy splits in $\mathcal{B}$ yield τ than yield any other candidate value. We use the convention that the maximum over an empty collection is zero.

Then,

$$\{\tau\} = \arg \max_{c \in \mathbb{R}} |\{S \in \mathcal{B} : \theta_S = c\}|. \quad (24)$$

The theorem states that τ can be identified even when we do not know the exact held-out proxy W_j , if it is the unique most frequent candidate value. The following corollary is the strict-majority special case of Theorem 2. It gives both the unique most frequent candidate value and the corresponding median aggregation result.

Corollary 1 (Target-valued majority and median aggregation). Suppose $\mathcal{B} \neq \emptyset$ and

$$|\{S \in \mathcal{B} : \theta_S = \tau\}| > \frac{|\mathcal{B}|}{2}. \quad (25)$$

That is, more than half of the retained proxy splits yield the true causal effect. Then

$$\{\tau\} = \arg \max_{c \in \mathbb{R}} |\{S \in \mathcal{B} : \theta_S = c\}|, \quad (26)$$

and

$$\text{median}((\theta_S)_{S \in \mathcal{B}}) = \tau. \quad (27)$$

Figure 2 illustrates the result. The labels $W_{1:3}$ and $W_{4:5}$ denote grouped proxy blocks, not single variables. Under Assumption 1, if all five singleton proxy splits pass the balance screen and proxy splits 1, 2, 3 satisfy the grouped versions of Assumptions 2 and 3, then three of the five singleton

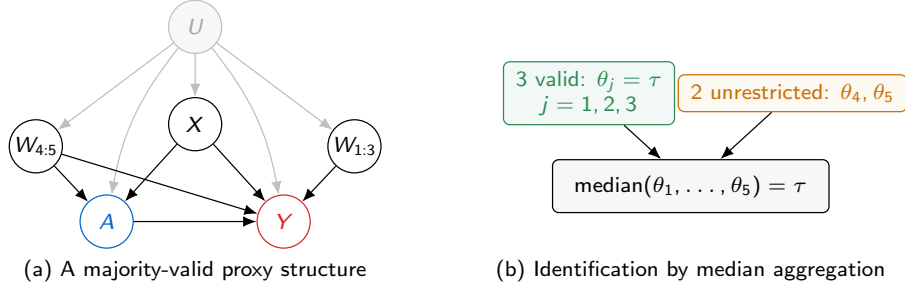


Figure 2: Median aggregation over five singleton proxy splits.

candidate values equal τ , so their median equals τ .

3.3. Approximate PROBE Balance and Bias Bound

Exact balance is an ideal endpoint. We now consider representations that achieve only approximate, rather than exact, balance. Fix a representation $\phi \in \Phi_j$ and write $T_j \triangleq (X, W_j)$; the notation Z_j^ϕ was defined at the beginning of this section. Define

$$e_{j,\phi}(z) \triangleq P(A = 1 \mid Z_j^\phi = z), \quad q_{j,\phi}(t, z) \triangleq P(A = 1 \mid T_j = t, Z_j^\phi = z). \quad (28)$$

Their difference is the treatment information that remains in the held-out variables after conditioning on the representation.

Definition 2 (Residual treatment discrepancy). Define

$$D_{\text{res},j}^2(\phi) \triangleq \mathbb{E}[\{q_{j,\phi}(T_j, Z_j^\phi) - e_{j,\phi}(Z_j^\phi)\}^2], \quad (29)$$

$$D_{\text{res},j}(\phi) \triangleq \{D_{\text{res},j}^2(\phi)\}^{1/2}. \quad (30)$$

The discrepancy measures the treatment information in T_j that remains after conditioning on Z_j^ϕ .

Proposition 2 (Zero discrepancy and exact balance).

$$D_{\text{res},j}(\phi) = 0 \iff A \perp\!\!\!\perp T_j \mid Z_j^\phi. \quad (31)$$

To relate the discrepancy with the causal bias bound, we define

$$P_{j,\phi,z} \triangleq \mathcal{L}(T_j \mid Z_j^\phi = z), \quad (32)$$

For $P_{Z_j^\phi}$ -almost every z and $a \in \{0, 1\}$, write $Q_{a,j,z}^\phi \triangleq \mathcal{L}(V_j, U \mid A = a, Z_j^\phi = z)$ and $\mu_{a,j,z}^\phi(v, u) \triangleq \mathbb{E}\{Y(a) \mid V_j = v, U = u, Z_j^\phi = z\}$. Let $\mathcal{Q}_{j,z}^\phi$ be the collection of probability laws Q on (V_j, U) such that $Q \ll \mathcal{L}(V_j, U \mid Z_j^\phi = z)$ and $\mathbb{E}_Q[\mu_{a,j,z}^\phi(V_j, U)] < \infty$ for both $a \in \{0, 1\}$. We assume the following:

Assumption 4 (L_2 outcome-relevant stability). For every representation ϕ under consideration, there is a finite scalar $\Gamma_j(\phi) > 0$ such that, for $P_{Z_j^\phi}$ -almost every z , every $a \in \{0, 1\}$, and all $Q, Q' \in \mathcal{Q}_{j,z}^\phi$,

$$\left| \mathbb{E}_Q\{\mu_{a,j,z}^\phi\} - \mathbb{E}_{Q'}\{\mu_{a,j,z}^\phi\} \right| \leq \Gamma_j(\phi) \left\| \frac{d(\mathcal{K}_j Q - \mathcal{K}_j Q')}{dP_{j,\phi,z}} \right\|_{L_2(P_{j,\phi,z})}. \quad (33)$$

If $\mathcal{K}_j Q = \mathcal{K}_j Q'$, the right-hand side is zero. Assumption 4 therefore implies Assumption 3.

Theorem 3 (Approximate PROBE bias bound). Suppose Assumptions 1, 2, and 4 hold for Z_j^ϕ . Suppose there is a constant $\eta \in (0, 1/2]$ such that $\eta \leq e_{j,\phi}(z) \leq 1 - \eta$ for $P_{Z_j^\phi}$ -almost every z . Then

$$\left| \mathbb{E}\left[\mathbb{E}(Y \mid A = 1, Z_j^\phi) - \mathbb{E}(Y \mid A = 0, Z_j^\phi)\right] - \tau \right| \leq \frac{\Gamma_j(\phi)}{\eta(1 - \eta)} D_{\text{res},j}(\phi). \quad (34)$$

The theorem implies the following sensitivity result.

Proposition 3 (Population sensitivity region). Suppose Assumptions 1, 2, and 4 hold for Z_j^ϕ , and suppose there is a constant $\eta \in (0, 1/2]$ such that $\eta \leq e_{j,\phi}(z) \leq 1 - \eta$ for $P_{Z_j^\phi}$ -almost every z . Define

$$\tau_\phi \triangleq \mathbb{E}\left[\mathbb{E}(Y \mid A = 1, Z_j^\phi) - \mathbb{E}(Y \mid A = 0, Z_j^\phi)\right]. \quad (35)$$

Then

$$\tau \in \left[\tau_\phi - \frac{\Gamma_j(\phi)}{\eta(1 - \eta)} D_{\text{res},j}(\phi), \tau_\phi + \frac{\Gamma_j(\phi)}{\eta(1 - \eta)} D_{\text{res},j}(\phi) \right]. \quad (36)$$

This is a population sensitivity region, not a finite-sample confidence interval. When $D_{\text{res},j}(\phi) = 0$, it collapses to the point-identified value in Theorem 1. The residual discrepancy is a functional of the observed distribution. The constant $\Gamma_j(\phi)$ is structural and must be justified by scientific knowledge, additional theory, or sensitivity analysis.

4. Representation Learning and AIPW Estimation

The identification results concern one fixed representation. We now select a representation from data and estimate the adjustment functional that it induces. The representation-learning and AIPW analysis below uses one prespecified held-out block j throughout.

4.1. Representation Learning

Recall that one observed record is $O_i = (X_i, W_i, A_i, Y_i)$. Partition the observed sample into three mutually disjoint subsets,

$$\mathcal{I}_D, \quad \mathcal{I}_N, \quad \mathcal{I}_E, \quad (37)$$

with respective sizes n_D , n_N , and n_E . The discrepancy-learning sample $\mathcal{I}_D$ selects a representation. The nuisance-learning sample $\mathcal{I}_N$ estimates the outcome regressions and propensity used by AIPW. The evaluation sample $\mathcal{I}_E$ calculates the final estimate. Conditional on the preceding samples, the objects evaluated on the next sample are fixed.

Let $\mathcal{F}_{j,n_D} \subseteq \Phi_j$ be a prespecified, possibly infinite class of deterministic encoders. For $\phi \in \mathcal{F}_{j,n_D}$, write $Z_j^\phi = \phi(V_j)$ and retain $T_j = (X, W_j)$ from Section 3.3.

Assumption 5 (Common representation overlap). There is a constant $\eta \in (0, 1/2]$ such that, for every $\phi \in \mathcal{F}_{j,n_D}$,

$$\eta \leq e_{j,\phi}(Z_j^\phi) \leq 1 - \eta \quad \text{almost surely.} \quad (38)$$

For any prespecified nonempty proper subset $S \subset \{1, \dots, J\}$, the results below also apply with (V_j, W_j, Z_j^ϕ) replaced by $(V_S, W_S, \phi(V_S))$ and with the same assumptions imposed on this grouped split.

The Brier-risk definition below estimates residual discrepancy by comparing squared-loss prediction of A from Z_j^ϕ alone with prediction from (T_j, Z_j^ϕ) (Brier, 1950).

Definition 3 (Brier-risk residual discrepancy). Let $\mathcal{H}_{0,n_D}$ and $\mathcal{H}_{1,n_D}$ be prespecified classes of measurable $[0, 1]$ -valued predictors with inputs Z_j^ϕ and (T_j, Z_j^ϕ) , respectively, fixed independently of $\mathcal{I}_D$. Assume $\mathcal{H}_{1,n_D}$ contains each member of $\mathcal{H}_{0,n_D}$ lifted to ignore T_j . For $b \in \{0, 1\}$, define

$$R_{0,\phi}(h) \triangleq \mathbb{E}[\{A - h(Z_j^\phi)\}^2], \quad \hat{R}_{0,\phi}(h) \triangleq \frac{1}{n_D} \sum_{i \in \mathcal{I}_D} \{A_i - h(Z_{j,i}^\phi)\}^2, \quad (39)$$

$$R_{1,\phi}(h) \triangleq \mathbb{E}[\{A - h(T_j, Z_j^\phi)\}^2], \quad \hat{R}_{1,\phi}(h) \triangleq \frac{1}{n_D} \sum_{i \in \mathcal{I}_D} \{A_i - h(T_{j,i}, Z_{j,i}^\phi)\}^2, \quad (40)$$

where $Z_{j,i}^\phi = \phi(V_{j,i})$. For each ϕ , let $\hat{h}_{b,\phi} \in \mathcal{H}_{b,n_D}$ be an approximate empirical minimizer satisfying, for some $\varepsilon_{b,\phi} \geq 0$,

$$\hat{R}_{b,\phi}(\hat{h}_{b,\phi}) \leq \inf_{h \in \mathcal{H}_{b,n_D}} \hat{R}_{b,\phi}(h) + \varepsilon_{b,\phi}, \quad b \in \{0, 1\}. \quad (41)$$

Define $\varepsilon_{\text{crit},j} \triangleq \sup_{\phi \in \mathcal{F}_{j,n_D}} (\varepsilon_{0,\phi} + \varepsilon_{1,\phi})$ and

$$\tilde{D}_{j,n_D}^2(\phi) \triangleq \max \left\{ (\hat{R}_{0,\phi}(\hat{h}_{0,\phi}) - \hat{R}_{1,\phi}(\hat{h}_{1,\phi})), 0 \right\}, \quad (42)$$

Each encoder ϕ and critic h give a squared-loss function of one observation: $\{A - h(\phi(V_j))\}^2$ or

$\{A - h(T_j, \phi(V_j))\}^2$. Varying both ϕ and h over their prespecified classes gives the loss classes $\mathcal{L}_{0,j,n_D}$ and $\mathcal{L}_{1,j,n_D}$, respectively. Let $\mathfrak{R}_{n_D}(\mathcal{L})$ denote the expected Rademacher complexity of a loss class $\mathcal{L}$. For $\delta_D \in (0, 1)$, define

$$r_{j,n_D}(\delta_D) \triangleq 2\{\mathfrak{R}_{n_D}(\mathcal{L}_{0,j,n_D}) + \mathfrak{R}_{n_D}(\mathcal{L}_{1,j,n_D})\} + 2\sqrt{\frac{\log(4/\delta_D)}{2n_D}}. \quad (43)$$

This radius controls sampling fluctuation uniformly over the encoder–critic choices. The complexity terms account for the flexibility of the two loss classes, and the last term sets the failure-probability level δ_D .

Proposition 4 (Uniform residual-discrepancy deviation). For each $\phi \in \mathcal{F}_{j,n_D}$, define

$$a_{0,n_D}(\phi) \triangleq \inf_{h \in \mathcal{H}_{0,n_D}} R_{0,\phi}(h) - \mathbb{E}[\{A - e_{j,\phi}(Z_j^\phi)\}^2], \quad (44)$$

$$a_{1,n_D}(\phi) \triangleq \inf_{h \in \mathcal{H}_{1,n_D}} R_{1,\phi}(h) - \mathbb{E}[\{A - q_{j,\phi}(T_j, Z_j^\phi)\}^2], \quad (45)$$

$$a_{j,n_D} \triangleq \sup_{\phi \in \mathcal{F}_{j,n_D}} \{a_{0,n_D}(\phi) + a_{1,n_D}(\phi)\}. \quad (46)$$

Under the measurability conditions stated in Assumption 6, with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_{j,n_D}} |\tilde{D}_{j,n_D}^2(\phi) - D_{\text{res},j}^2(\phi)| \leq r_{j,n_D}(\delta_D) + a_{j,n_D} + \varepsilon_{\text{crit},j}. \quad (47)$$

At any fixed confidence level $1 - \delta_D$, this error bound is $O(n_D^{-1/2})$ if both Rademacher complexities, a_{j,n_D} , and $\varepsilon_{\text{crit},j}$ are each $O(n_D^{-1/2})$.

Definition 4 (Approximate empirical minimizer). For a prescribed tolerance $\varepsilon_{\text{enc},j} \geq 0$, an approximate empirical minimizer is an encoder $\hat{\phi}_j \in \mathcal{F}_{j,n_D}$ selected measurably from the training observations and algorithmic seed such that

$$\tilde{D}_{j,n_D}^2(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_{j,n_D}} \tilde{D}_{j,n_D}^2(\phi) + \varepsilon_{\text{enc},j}. \quad (48)$$

An exact empirical minimizer has $\varepsilon_{\text{enc},j} = 0$.

Proposition 5 (Residual-discrepancy oracle inequality). Let $\delta_D \in (0, 1)$. Suppose the pre-specified encoder and $[0, 1]$ -valued critic classes, approximate critic fits in Proposition 4 and Assumption 6 hold, and $\hat{\phi}_j$ satisfies Equation (48). Then, with probability at least $1 - \delta_D$,

$$D_{\text{res},j}^2(\hat{\phi}_j) \leq \inf_{\phi \in \mathcal{F}_{j,n_D}} D_{\text{res},j}^2(\phi) + 2\{r_{j,n_D}(\delta_D) + a_{j,n_D} + \varepsilon_{\text{crit},j}\} + \varepsilon_{\text{enc},j}. \quad (49)$$

Condition on the training observations $\{O_i : i \in \mathcal{I}_D\}$ and any algorithmic seed used in training, which fixes $\hat{\phi}_j$. Here and below, population expectations involving this learned encoder use a fresh observation $O = (X, W, A, Y)$ independent of the training observations and seed, with this conditioning understood. The learned representation and its population adjustment functional are

$$\begin{aligned} Z_j^{\hat{\phi}_j} &\triangleq \hat{\phi}_j(V_j), & \hat{Z}_j &\triangleq Z_j^{\hat{\phi}_j}, \\ \tau_{\hat{Z}_j} &\triangleq \mathbb{E}[\mathbb{E}(Y \mid A = 1, \hat{Z}_j) - \mathbb{E}(Y \mid A = 0, \hat{Z}_j)]. \end{aligned} \quad (50)$$

Thus $\tau_{\hat{Z}_j} = \tau_\phi|_{\phi=\hat{\phi}_j}$, where τ_ϕ is the population adjustment functional defined in Proposition 3. The adjustment functional is used whenever its two conditional means are absolutely integrable under the law of $\hat{Z}_j$.

Theorem 4 (Causal error of the learned representation). Let $\delta_D \in (0, 1)$. Suppose the prespecified encoder and $[0, 1]$ -valued critic classes, approximate critic fits in Proposition 4 and Assumption 6 hold, and $\hat{\phi}_j$ satisfies Equation (48). Suppose Assumptions 1, 2, and 4 hold for every Z_j^ϕ with $\phi \in \mathcal{F}_{j, n_D}$, and there is a constant $\eta \in (0, 1/2]$ such that $\eta \leq e_{j, \phi}(Z_j^\phi) \leq 1 - \eta$ almost surely for every such ϕ . Then, with probability at least $1 - \delta_D$,

$$|\tau_{\hat{Z}_j} - \tau| \leq \frac{\Gamma_j(\hat{\phi}_j)}{\eta(1 - \eta)} \left[\inf_{\phi \in \mathcal{F}_{j, n_D}} D_{\text{res}, j}^2(\phi) + 2\{r_{j, n_D}(\delta_D) + a_{j, n_D} + \varepsilon_{\text{crit}, j}\} + \varepsilon_{\text{enc}, j} \right]^{1/2}. \quad (51)$$

At fixed confidence and fixed overlap constant η , the right-hand side is $O_p(n_D^{-1/4})$ if $\inf_{\phi \in \mathcal{F}_{j, n_D}} D_{\text{res}, j}^2(\phi) = 0$, each of $r_{j, n_D}(\delta_D)$, a_{j, n_D} , $\varepsilon_{\text{crit}, j}$, and $\varepsilon_{\text{enc}, j}$ is $O(n_D^{-1/2})$, and $\Gamma_j(\hat{\phi}_j) = O_p(1)$. This is a conditional upper-bound rate; it is not a universal or minimax rate.

4.2. AIPW Estimation

This section estimates the adjustment functional $\tau_{\hat{Z}_j}$ of the learned representation with an augmented inverse probability weighted (AIPW) estimator (Robins et al., 1994; Tsiatis, 2006) computed on the evaluation sample. Define the conditional outcome means and representation propensity

$$m_{a, j}(z) \triangleq \mathbb{E}(Y \mid A = a, \hat{Z}_j = z), \quad (52)$$

$$e_j(z) \triangleq P(A = 1 \mid \hat{Z}_j = z). \quad (53)$$

For the observed adjustment functional $\tau_{\hat{Z}_j}$, the following is the AIPW score:

$$\text{IF}_j(O) \triangleq m_{1, j}(\hat{Z}_j) - m_{0, j}(\hat{Z}_j) + \frac{A\{Y - m_{1, j}(\hat{Z}_j)\}}{e_j(\hat{Z}_j)} - \frac{(1 - A)\{Y - m_{0, j}(\hat{Z}_j)\}}{1 - e_j(\hat{Z}_j)} - \tau_{\hat{Z}_j}. \quad (54)$$

When this score is square-integrable, it is the *efficient influence function* in the nonparametric model for $(\hat{Z}_j, A, Y)$. Bounded outcomes together with uniform representation overlap provide one sufficient condition for square integrability. Use the nuisance-learning sample, with $\hat{Z}_{j, i} = \hat{\phi}_j(V_{j, i})$, to estimate the nuisance functions. Denote the estimates by $\hat{m}_{0, j}$, $\hat{m}_{1, j}$, and $\hat{e}_j$.

Definition 5 (Honest AIPW estimator). For $i \in \mathcal{I}_E$, define $\hat{Z}_{j, i} = \hat{\phi}_j(V_{j, i})$ and

$$\text{UIF}_j(O_i) \triangleq \hat{m}_{1, j}(\hat{Z}_{j, i}) - \hat{m}_{0, j}(\hat{Z}_{j, i})$$

$$\begin{aligned}
& + \frac{A_i\{Y_i - \hat{m}_{1,j}(\hat{Z}_{j,i})\}}{\hat{e}_j(\hat{Z}_{j,i})} \\
& - \frac{(1 - A_i)\{Y_i - \hat{m}_{0,j}(\hat{Z}_{j,i})\}}{1 - \hat{e}_j(\hat{Z}_{j,i})}.
\end{aligned} \tag{55}$$

The score UIF_j is the uncentered plug-in version of IF_j : it replaces the nuisance functions by their estimates and omits the centering term $\tau_{\hat{Z}_j}$. The honest AIPW estimator is

$$\hat{\tau}_j^{\text{AIPW}} \triangleq \frac{1}{n_E} \sum_{i \in \mathcal{I}_E} \text{UIF}_j(O_i). \tag{56}$$

The plug-in score has the doubly robust remainder structure of augmented inverse probability weighting (Chernozhukov et al., 2017; Kennedy, 2022): its conditional bias is a product of the propensity error and the outcome-regression errors, as the next proposition states exactly.

Proposition 6 (Exact signed nuisance remainder). Conditional on $\mathcal{I}_D$ and $\mathcal{I}_N$,

$$\mathbb{E}\{\text{UIF}_j(O)\} - \tau_{\hat{Z}_j} = \mathbb{E}\left[\{\hat{e}_j(\hat{Z}_j) - e_j(\hat{Z}_j)\} \left\{ \frac{\hat{m}_{1,j}(\hat{Z}_j) - m_{1,j}(\hat{Z}_j)}{\hat{e}_j(\hat{Z}_j)} + \frac{\hat{m}_{0,j}(\hat{Z}_j) - m_{0,j}(\hat{Z}_j)}{1 - \hat{e}_j(\hat{Z}_j)} \right\}\right]. \tag{57}$$

The remainder is zero if the propensity estimator is correct. It is also zero if both conditional outcome estimators are correct.

Clip the propensity estimate so that $\eta \leq \hat{e}_j(z) \leq 1 - \eta$. Conditional on the two learning samples, that is, on their observations and the seeds used to fit the encoder and the nuisance functions, define for a function g of $\hat{Z}_j$ the norm $\|g\|_{2,\hat{Z}_j} \triangleq [\mathbb{E}\{g(\hat{Z}_j)^2 \mid \mathcal{I}_D, \mathcal{I}_N\}]^{1/2}$ and the realized nuisance errors

$$q_{e,j} \triangleq \|\hat{e}_j - e_j\|_{2,\hat{Z}_j}, \quad q_{a,j} \triangleq \|\hat{m}_{a,j} - m_{a,j}\|_{2,\hat{Z}_j}, \quad a \in \{0, 1\}. \tag{58}$$

Let $\sigma_j^2 \triangleq \text{Var}\{\text{UIF}_j(O) \mid \mathcal{I}_D, \mathcal{I}_N\}$ denote the conditional variance of the plug-in score; it is finite whenever $\mathbb{E}(Y^2) < \infty$ and the outcome estimates are square-integrable. For $\delta_E \in (0, 1)$, define

$$\varepsilon_{\text{AIPW},j} \triangleq \frac{\sigma_j}{\sqrt{n_E \delta_E}} + \frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \tag{59}$$

The next theorem is the final guarantee of this section: the causal error of the honest AIPW estimator is at most the estimation radius $\varepsilon_{\text{AIPW},j}$ plus the stability-weighted residual discrepancy of the learned representation.

Theorem 5 (Finite-sample causal error of honest AIPW). Suppose $\sigma_j < \infty$. Conditional on the two learning samples, with probability at least $1 - \delta_E$ over $\mathcal{I}_E$,

$$|\hat{\tau}_j^{\text{AIPW}} - \tau_{\hat{Z}_j}| \leq \varepsilon_{\text{AIPW},j}. \tag{60}$$

If, in addition, the assumptions of Theorem 4 hold, then on the same event

$$|\hat{\tau}_j^{\text{AIPW}} - \tau| \leq \varepsilon_{\text{AIPW},j} + \frac{\Gamma_j(\hat{\phi}_j)}{\eta(1-\eta)} D_{\text{res},j}(\hat{\phi}_j). \quad (61)$$

Corollary 2 (Consistency). Consider a sequence of encoder and critic classes, sample splits, nuisance estimators, optimization tolerances, and selected encoders $\hat{\phi}_{j,n}$ indexed by the total sample size n , with a common overlap constant $\eta \in (0, 1/2]$, such that Theorem 5 applies at every n . Suppose the following three conditions hold.

- (C1) *Vanishing evaluation error.* $\delta_{\text{E},n} \rightarrow 0$ and $\sigma_{j,n}^2/(n_{\text{E}}\delta_{\text{E},n}) \rightarrow 0$. The evaluation sample grows faster than the confidence level shrinks.
- (C2) *Vanishing nuisance product.* $q_{e,j,n}(q_{0,j,n} + q_{1,j,n}) \xrightarrow{p} 0$. The propensity and outcome-regression errors need not vanish individually; their product vanishes, for example, when each is $o_p(n^{-1/4})$.
- (C3) *Vanishing learned imbalance.* $\Gamma_j(\hat{\phi}_{j,n})D_{\text{res},j}(\hat{\phi}_{j,n}) \xrightarrow{p} 0$. By Proposition 5, this holds when $\inf_{\phi \in \mathcal{F}_{j,n_{\text{D}}}} D_{\text{res},j}^2(\phi)$, the two Rademacher complexities, $a_{j,n_{\text{D}}}$, $\varepsilon_{\text{crit},j,n}$, $\varepsilon_{\text{enc},j,n}$, and $\delta_{\text{D},n}$ all converge to zero and $\Gamma_j(\hat{\phi}_{j,n}) = O_p(1)$.

Then $\hat{\tau}_{j,n}^{\text{AIPW}} \xrightarrow{p} \tau$.

Equation (61) separates the final error into an AIPW estimation term and a hidden-confounding sensitivity term. The selected stability constant $\Gamma_j(\hat{\phi}_j)$, critic approximation error $a_{j,n_{\text{D}}}$, optimization tolerances, and nuisance errors are not generally observed. The result becomes an operational numerical certificate after these quantities are bounded by theory, computation, external information, or a declared sensitivity analysis.

4.3. Searching Across Held-Out Orientations

Exhaustive evaluation of all orientations in $\mathfrak{S}$ may be computationally expensive. We therefore consider randomized subsampling of the population candidates defined in Section 3.2. This result concerns computational search over exact population values. It is not a finite-sample clustering or causal-inference guarantee.

Let $T \triangleq |\mathfrak{S}| = 2^J - 2$ and $N \triangleq |\mathcal{B}|$. Let the search budget $M \in \{0, \dots, T\}$ be chosen independently of the observed data and all representation-learning outcomes. Conditional on $M = m$, sample m orientations uniformly without replacement from $\mathfrak{S}$. Let $\mathcal{R}_m$ be the sampled orientations that also belong to $\mathcal{B}$, and let $R \triangleq |\mathcal{R}_m|$. Then

$$R \mid M = m \sim \text{Hypergeometric}(T, N, m). \quad (62)$$

Conditional on $(M = m, R = r)$, the retained set $\mathcal{R}_m$ is a uniformly sampled r -element subset of $\mathcal{B}$.

For $c \in \mathbb{R}$, define the population count

$$n(c) \triangleq |\{S \in \mathcal{B} : \theta_S = c\}|. \quad (63)$$

Let

$$L \triangleq |\{c \in \mathbb{R} : c \neq \tau, n(c) > 0\}| \quad (64)$$

be the number of non-target population values. When $L \geq 1$, define

$$p_\tau \triangleq \frac{n(\tau)}{N}, \quad p_\star \triangleq \max_{c \neq \tau: n(c) > 0} \frac{n(c)}{N}, \quad \Delta \triangleq p_\tau - p_\star. \quad (65)$$

Theorem 6 (Recovery by random orientation search). Suppose $\mathcal{B} \neq \emptyset$ and

$$n(\tau) > \max_{c \neq \tau: n(c) > 0} n(c),$$

where the maximum over an empty collection is zero. Fix $m \in \{0, \dots, T\}$ such that $P(M = m) > 0$, and fix an integer $1 \leq r_0 \leq \min(m, N)$. Suppose

$$P(R \geq r_0 \mid M = m) \geq 1 - \eta_R(m). \quad (66)$$

The procedure returns failure when $R = 0$. If $L \geq 1$, select the value having the largest retained count and treat any tie as failure. Conditional on $M = m$, this procedure returns τ with probability at least

$$\left[1 - \eta_R(m) - L \exp \left\{ -\frac{r_0 \Delta^2}{2} \right\} \right]_+, \quad [x]_+ \triangleq \max(x, 0). \quad (67)$$

If $L = 0$, every retained candidate equals τ , so recovery succeeds whenever $R \geq 1$; neither p_τ , $p_\star$, nor Δ is needed in this case.

The probability in Theorem 6 concerns only the randomized computational search. If M is random and $P(R \geq r_0) \geq 1 - \eta_R$, averaging the same argument over M gives the unconditional version of equation (67) with $\eta_R(m)$ replaced by η_R . When $L = 0$, the success probability is at least $1 - \eta_R$ provided $r_0 \geq 1$.

5. Experiments

5.1. Questions, Comparators, and Reporting Rules

The experiments answer three separate questions. First, does proxy-supervised representation adjustment reduce ATE error relative to adjustment using X alone? Second, do the exact-assumption designs recover the known balancing structure? Third, does the method show a useful descriptive signal in an observational benchmark with a noisy randomized reference? We report mean absolute error, root mean squared error, paired wins, held-out discrepancy, overlap diagnostics, and Monte Carlo uncertainty. Oracle procedures that observe U or the true balancing score are simulation

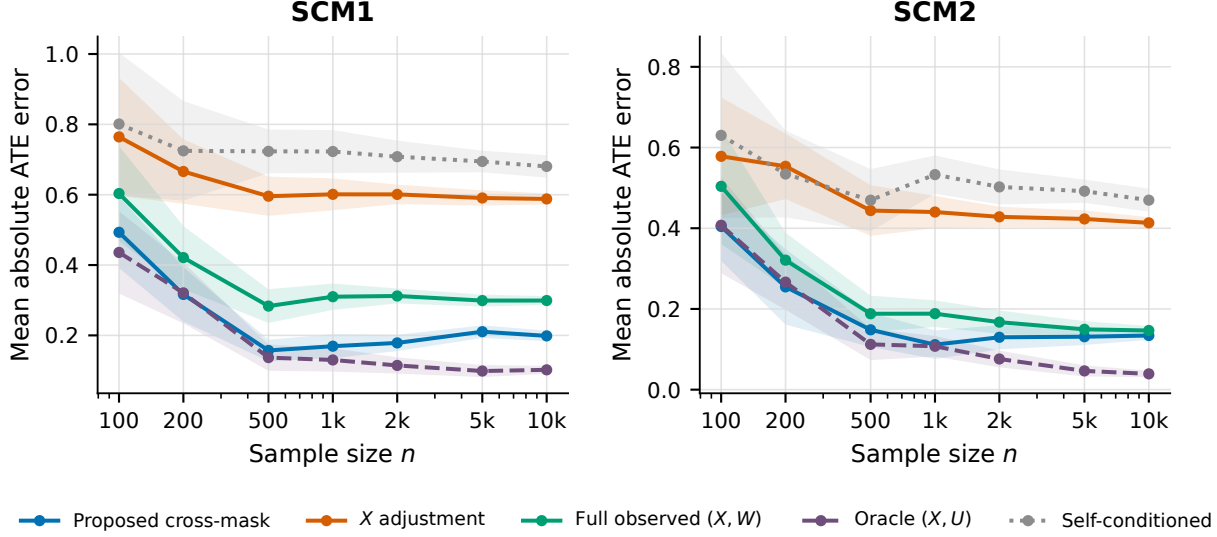


Figure 3: Mean absolute ATE error across twenty data sets; bands show approximate 95% Monte Carlo intervals for the mean.

references, not implementable competitors. The theory in Sections 3 and 4 applies to one prespecified held-out block at a time. Experimental aggregation across several orientations is reported as an empirical extension and is not presented as a separate identification theorem.

5.2. Honest Approximate SCMs

The primary approximate study fixes the causal order $U \rightarrow (X, W)$, $(X, U) \rightarrow A$, and $(A, X, U) \rightarrow Y$. Both SCMs use $\dim(X) = 20$ and $\dim(W) = 10$, with twenty independently generated data sets and nested prefixes $n \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$. SCM1 uses linear proxy measurements and a nonlinear outcome component; SCM2 changes the structural equations to nonlinear transformations while preserving the same causal order and an informative monotone proxy channel. The proposed cross-mask method learns a representation from X and all but one proxy coordinate, audits balance with the held-out coordinate, and averages over masks.

At $n = 1000$, the mean absolute errors of proposed/ X -adjustment/full- (X, W) /oracle- (X, U) were 0.169/0.601/0.310/0.130 in SCM1 and 0.112/0.440/0.188/0.107 in SCM2. At $n = 10000$, the corresponding proposed errors were 0.199 and 0.134, versus 0.588 and 0.413 for X adjustment. These are finite-sample simulation results, not a proof of the audit-view identification theorem. The self-conditioned heuristic remained substantially biased, which is consistent with the self-conditioning warning in Section 3.

5.3. Exact-Assumption Tracks A and B

Track A uses a finite balancing score, $\dim(X) = 20$, nine representation-side proxy coordinates, and one independent audit coordinate. The DGP enforces strict population overlap, exact decoding of the discrete confounding state from the representation-side view, and an injective Gaussian audit

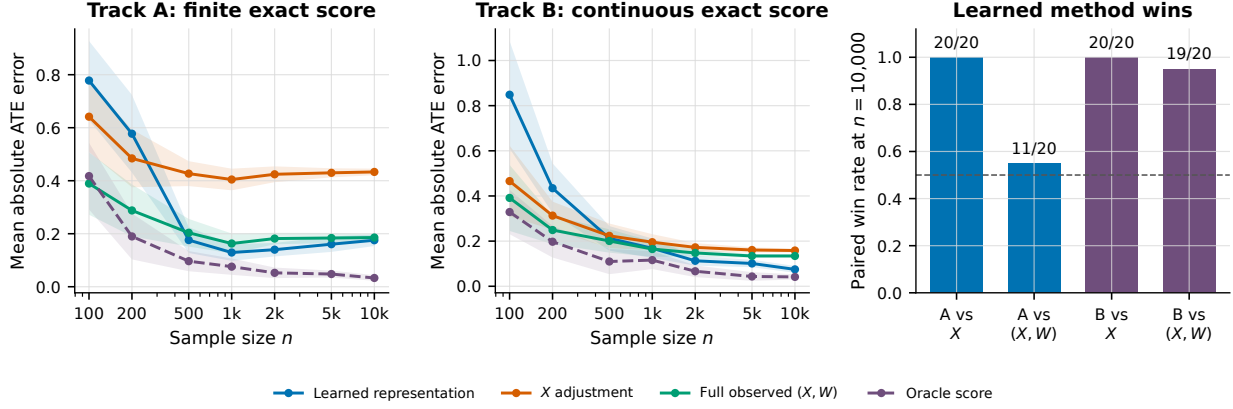


Figure 4: Mean absolute ATE error and paired win rates for exact-assumption Tracks A and B over twenty replications.

channel. At $n = 10000$ over twenty replications, mean absolute error was 0.175 for the learned audit representation, 0.433 for X adjustment, 0.186 for direct full-observed adjustment, and 0.033 for the oracle balancing score. The primary comparison with X passed, while the prespecified secondary superiority criterion against full-observed adjustment failed.

Track B uses a continuous exact balancing score and a linear representation class matched to that score. At $n = 10000$, mean absolute error was 0.0748 for the learned representation, 0.1583 for fitted X adjustment, and 0.1342 for fitted full-observed adjustment. The learned method beat X in 20/20 and full-observed adjustment in 19/20 paired replications. Its mean rank-recovery Spearman correlation was 0.915. The held-out conditional-dependence diagnostic used finite random features and is a surrogate, not an exact population KCI test; the design also covers one prespecified scalar- U SCM family.

5.4. WSC Observational Benchmark

The WSC study forms an observational subset of 1,105 participants whose randomized mathematics assignment agrees with pretreatment preference, and targets the population ATE in the original 2,200 randomized participants. The full randomized difference in means is 0.760 with standard error 0.151 and serves only as a noisy benchmark. Across ten algorithmic split repetitions on this one fixed data set, the unconstrained cross-mask estimate averaged 1.302 with split standard deviation 0.065 and absolute benchmark gap 0.541, compared with gaps 2.690 for the naive observational contrast, 1.704 for direct X adjustment, and 0.672 for direct (X, W) adjustment. The unconstrained cross-mask method is the proposed WSC method shown in Figure 5. Its practical-overlap gate passed 0/10 repetitions. The benchmark therefore supplies descriptive evidence only, not a validated causal-performance claim, causal ground truth, or a test of proxy validity.

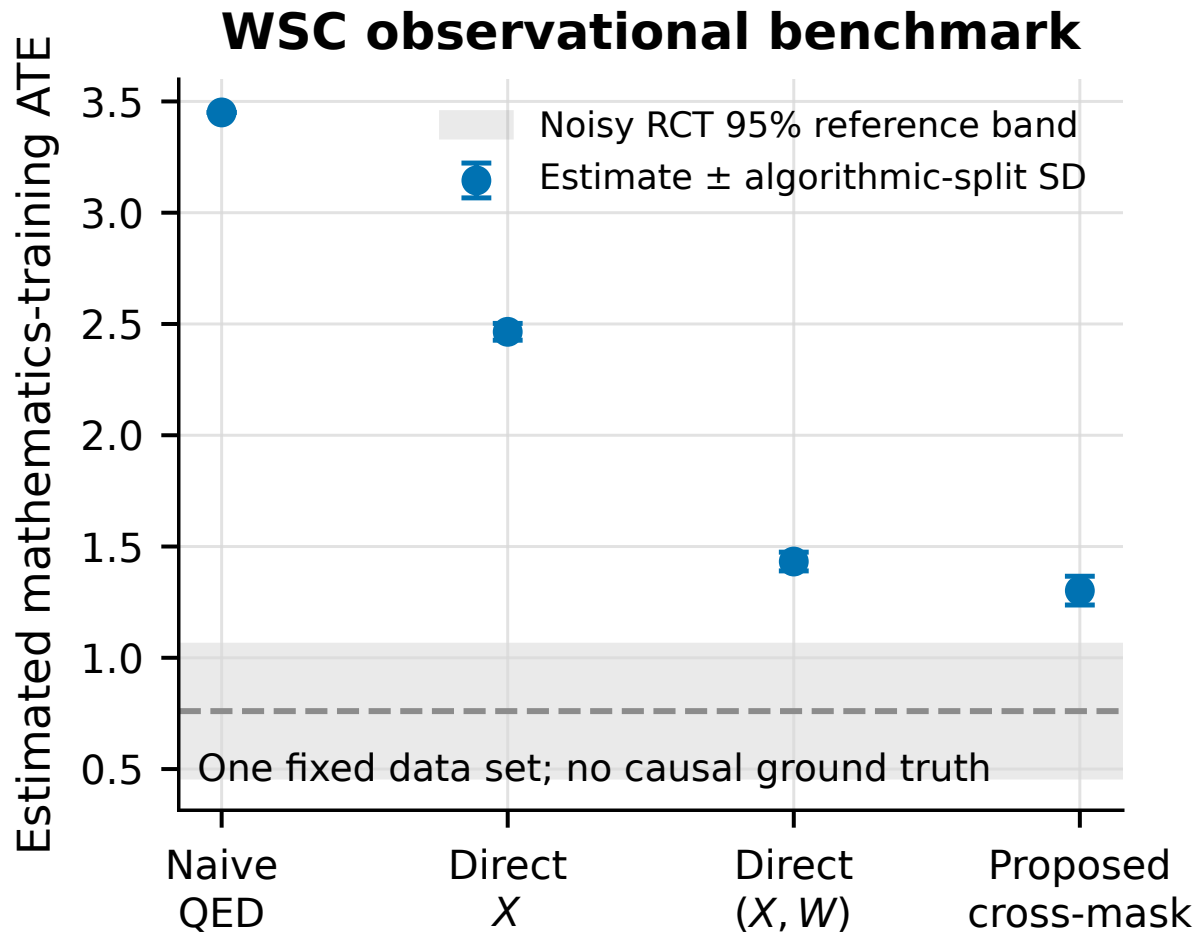


Figure 5: WSC estimates relative to the randomized reference; error bars show standard deviations across ten algorithmic splits.

6. Discussion and Limitations

6.1. What the Method Delivers

For one prespecified block j , the canonical method uses W_{-j} to construct an outcome-free representation and uses the held-out block W_j to measure the residual imbalance of (X, W_j) across treatment arms within that representation. Under uniform outcome-relevant stability, the product $\Gamma_j D_j(\phi_j)$ is a sensitivity certificate for hidden-confounding bias. This is stronger than reporting a conditional-independence test alone because it targets a causal error scale, but weaker than point identification when $D_j(\phi_j) > 0$ because Γ_j is not generally identified from the observed distribution.

6.2. Testable and Structural Components

Held-out joint balance, empirical overlap, and minimum stratum mass are observable properties. Joint latent exchangeability, separation of the held-out proxy block, outcome-relevant completeness, and outcome-relevant stability are structural assumptions about the full-data law. Exact balance is therefore an observable population condition that can only be assessed up to sampling error, while its causal interpretation still requires structural assumptions. A large held-out discrepancy can falsify exact balance; a small test statistic cannot prove exact balance, completeness, or stability.

6.3. Attainability and Realized Bias

The exact identification theorem is conditional on the existence of a representation $Z_j = \phi_j(V_j)$ that attains exact held-out proxy balance. The theorem does not guarantee such attainability in a restricted representation class. In the approximate regime, reducing $D_j(\phi_j)$ reduces the certified worst-case bias under a common stability budget, but it need not reduce the realized bias when candidate representations have different stability constants or when the structural stability condition fails.

6.4. Learning and Inference Gaps

The current explicit finite-sample theorem covers finite candidate libraries with hard finite strata and fixed-fold evaluation. The neural experiments use soft assignments and covariance-weighted surrogate objectives that are not algebraically identical to the canonical MMD aggregation. A neural-class generalization bound, a soft-to-hard guarantee, and valid joint inference after adaptive representation selection remain open. These gaps prevent the manuscript from describing every neural result as a direct verification of the theory, even when its empirical error is lower.

6.5. Audit Views and Practical Scope

PROBE uses one prespecified proxy block for held-out evaluation and the remaining high-dimensional measurement for representation learning. This design blocks literal self-copying of the evaluated block. It does not by itself establish the required held-out-block structure, so point

identification still needs the structural assumptions in Section 3. The experiments repeat the procedure across several held-out orientations and aggregate their estimates as an empirical extension. The population median result in Corollary 1 justifies aggregation when the target-valued cluster contains strictly more than half of the balance-screened orientations. A strict majority of structurally valid orientations is sufficient, but not necessary; the current finite-sample learning and inference theory remains fixed- j . The WSC experiment further shows that an encouraging benchmark gap can coexist with a failed practical-overlap rule. Real-data evidence must therefore report overlap, held-out discrepancies, and benchmark uncertainty without treating any one diagnostic as ground truth.

6.6. Scope of the Contribution

The defensible contribution is the combination of a blockwise proxy-supervised representation, an observable conditional discrepancy, an exact identification theorem, a stability-indexed approximate guarantee, and an honest finite-sample estimation analysis. The paper does not claim that one generic correlated proxy point-identifies the ATE, that every smaller discrepancy has smaller realized bias without a common stability budget, or that the current neural optimizer reaches the population optimum. A systematic literature audit and broader external validation are required before any firstness or universality claim.

7. Conclusion

Single proxy balancing asks what a structured high-dimensional pretreatment proxy can reveal about hidden-confounding bias after learning a low-dimensional adjustment representation. For one prespecified held-out block j , PROBE learns $Z_j = \phi_j(V_j)$ and evaluates treatment balance through (X, W_j) . Exact held-out proxy balance identifies the ATE under the stated structural assumptions, while approximate balance yields the sensitivity certificate

$$|\mathbb{E}\{\mathbb{E}(Y \mid A = 1, Z_j) - \mathbb{E}(Y \mid A = 0, Z_j)\} - \tau| \leq \Gamma_j D_j(\phi_j).$$

Empirical discrepancy minimization and honest AIPW estimation give an end-to-end bound that separates learning, hidden-confounding, nuisance, and evaluation errors. The current simulations support the method in exact and approximate designs while exposing limitations from unattainable balance, weak proxy channels, surrogate mismatch, and practical-overlap loss. Neural-class generalization, soft-to-hard transfer, stability calibration, and external validation remain necessary for a submission-ready paper.

References

- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2017). Double/debiased machine learning for treatment and causal parameters. *arXiv preprint arXiv:1608.00060*.
- Cui, Y., Pu, H., Shi, X., Miao, W., and Tchetgen Tchetgen, E. (2024). Semiparametric proximal causal inference. *Journal of the American Statistical Association*, 119(546):1348–1359.
- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13:723–773.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Kennedy, E. H. (2022). Semiparametric doubly robust targeted double machine learning: A review. *arXiv preprint arXiv:2203.06469*.
- Lipsitch, M., Tchetgen Tchetgen, E., and Cohen, T. (2010). Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388.
- Miao, W., Geng, Z., and Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4):987–993.
- Park, C., Richardson, D. B., and Tchetgen Tchetgen, E. J. (2024). Single proxy control. *Biometrics*, 80(2):ujae027.
- Robins, J. M., Rotnitzky, A., and Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89(427):846–866.
- Rosenbaum, P. R. and Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55.
- Shalit, U., Johansson, F. D., and Sontag, D. (2017). Estimating individual treatment effect: Generalization bounds and algorithms. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3076–3085.
- Tchetgen Tchetgen, E. J., Ying, A., Cui, Y., Shi, X., and Miao, W. (2024). An introduction to proximal causal inference. *Statistical Science*, 39(3):375–390.
- Tsiatis, A. A. (2006). *Semiparametric Theory and Missing Data*. Springer Series in Statistics. Springer, New York.

A. Simulation Details

The existing project-local experiments will be documented by recording each data-generating process, parameter value, sample size, replication count, and random seed policy. Every manuscript figure and table will be linked to the script and machine-readable result from which it was produced.

The simulation audit will separate observed quantities from oracle quantities. In particular, any result that uses U or a direction constructed from U will be labeled as an oracle diagnostic and will not be presented as the output of a learnable procedure.

B. Proofs

B.1. Proofs for Section 3

Exact identification.

Proof of Theorem 1. Because $Z_j = \phi_j(V_j)$ is a measurable function of V_j , conditioning on (U, V_j, Z_j) is equivalent to conditioning on (U, V_j) . The two clauses of Assumption 2 therefore remain valid after additionally conditioning on Z_j . Thus, for each $a \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A, V_j, U, Z_j\} = \mathbb{E}\{Y(a) \mid V_j, U, Z_j\}, \quad (68)$$

$$\mathcal{L}(W_j \mid A, V_j, U, Z_j) = K_j(\cdot \mid V_j, U). \quad (69)$$

Integrating the second identity over $Q_{b,j,z}$ and using the definition of $\mathcal{K}_j$ gives

$$\mathcal{L}(X, W_j \mid A = b, Z_j = z) = \mathcal{K}_j Q_{b,j,z}. \quad (70)$$

Exact balance in equation (15) therefore implies

$$\mathcal{K}_j Q_{1,j,z} = \mathcal{K}_j Q_{0,j,z} \quad (71)$$

for P_{Z_j} -almost every z . Representation overlap gives $Q_{b,j,z} \ll \mathcal{L}(V_j, U \mid Z_j = z)$ for $b \in \{0, 1\}$. Assumption 1, equation (68), and Jensen's inequality give the required integrability, so $Q_{b,j,z} \in \mathcal{Q}_{j,z}$ almost everywhere. Assumption 3 therefore gives, for each $a \in \{0, 1\}$,

$$\mathbb{E}_{Q_{1,j,z}}\{\mu_{a,j,z}(V_j, U)\} = \mathbb{E}_{Q_{0,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (72)$$

By equation (68) and iterated expectation, for each $b \in \{0, 1\}$,

$$\mathbb{E}\{Y(a) \mid A = b, Z_j = z\} = \mathbb{E}_{Q_{b,j,z}}\{\mu_{a,j,z}(V_j, U)\}. \quad (73)$$

Equation (72) shows that the conditional mean of $Y(a)$ given (A, Z_j) does not depend on A . Consistency then gives

$$\begin{aligned} \mathbb{E}(Y \mid A = a, Z_j = z) &= \mathbb{E}\{Y(a) \mid A = a, Z_j = z\} \\ &= \mathbb{E}\{Y(a) \mid Z_j = z\}. \end{aligned} \quad (74)$$

Integrating over P_{Z_j} yields equation (16). $\square$

Non-nesting of canonical proximal and PROBE sufficient conditions.

Proof of Proposition 1. We give one finite-state law for each direction. In both constructions, X is constant, $Y = Y(A)$, and every treatment draw is generated by an exogenous randomizer independent of all proxy noises and outcome noises.

First, we construct a law satisfying the canonical proximal conditions but not the PROBE conditions for the designated orientation. Let

$$U \sim \text{Bernoulli}(1/2), \quad P(A = 1 \mid U = u) = \frac{1}{4} + \frac{1}{2}u,$$

and let

$$Z^{\text{tr}} = U \oplus E_Z, \quad Z^{\text{out}} = W_j = U \oplus E_W,$$

where E_Z and E_W are mutually independent Bernoulli(1/4) variables, independent of the treatment randomizer. Set $Y(a) = a + U$ and $V_j = Z^{\text{tr}}$. Consistency, integrability, and strict latent overlap hold. Independence of the exogenous noises gives the three canonical proxy-validity conditions.

The function

$$h(z^{\text{out}}, a) = a + 2z^{\text{out}} - \frac{1}{2}$$

is an outcome bridge because $\mathbb{E}(Z^{\text{out}} \mid U = u) = 1/4 + u/2$, so $\mathbb{E}\{h(Z^{\text{out}}, a) \mid U = u\} = a + u$. Iterated expectation gives the required observed bridge equation. The binary channel from U to Z^{tr} has matrix

$$\begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}$$

and determinant 1/2. Conditional on either treatment arm, both latent states have positive probability, and the corresponding unnormalized conditional-expectation matrix has determinant 3/32. It therefore has full column rank, which proves the canonical treatment-proxy completeness condition.

Now consider any measurable representation $Z_j = \phi_j(V_j)$. The Markov relation $U \rightarrow V_j \rightarrow Z_j$ holds. Conditional on (U, Z_j) , treatment and W_j are independent, and

$$\mathbb{E}(A \mid U, Z_j) = \mathbb{E}(W_j \mid U, Z_j) = \frac{1}{4} + \frac{1}{2}U.$$

The conditional covariance decomposition gives

$$\text{Cov}(A, W_j \mid Z_j) = \frac{1}{4} \text{Var}(U \mid Z_j).$$

Because $P(U = 1 \mid V_j = 0) = 1/4$ and $P(U = 1 \mid V_j = 1) = 3/4$,

$$P(U = 1 \mid Z_j) = \mathbb{E}\{P(U = 1 \mid V_j) \mid Z_j\} \in [1/4, 3/4]$$

almost surely. Hence $\text{Var}(U \mid Z_j) \geq 3/16$ and

$$\text{Cov}(A, W_j \mid Z_j) \geq \frac{3}{64} > 0.$$

Exact balance therefore fails for every such representation. This proves the first direction.

Second, we construct a law satisfying the PROBE conditions for the designated orientation but not the canonical proximal conditions. Let $U = (U_1, U_2)$, where U_1 and U_2 are independent Bernoulli(1/2) variables. Let Z^{tr} and Z^{out} be separate observed coordinates that agree almost surely with U_1 , and set

$$V_j = Z^{\text{tr}} = U_1, \quad W_j = Z^{\text{out}} = U_1, \quad Z_j = U_1.$$

Define

$$P(A = 1 \mid U_1, U_2) = \frac{3}{20} + \frac{1}{5}U_1 + \frac{2}{5}U_2,$$

and let ξ be an independent mean-zero Rademacher variable, also independent of the treatment randomizer. Set

$$Y(a) = a + U_1 + (1 + U_2)\xi.$$

The four latent treatment probabilities are 3/20, 11/20, 7/20, and 15/20, so strict latent overlap holds. Latent exchangeability follows from the independent outcome noise, and the common held-out channel holds because W_j is deterministic given U . Moreover,

$$P(A = 1 \mid Z_j = 0) = \frac{7}{20}, \quad P(A = 1 \mid Z_j = 1) = \frac{11}{20},$$

so representation overlap holds. Because $W_j = Z_j$ and X is constant, exact balance also holds.

It remains to verify the all- Q outcome-relevant completeness condition. Within a stratum $Z_j = z$, every admissible law $Q \in \mathcal{Q}_{j,z}$ is supported on $V_j = U_1 = z$, while U_2 may vary. On this support,

$$\mu_{a,j,z}(v, u) = a + z.$$

Thus, for any admissible Q, Q' and either potential treatment level,

$$\mathcal{K}_j Q = \mathcal{K}_j Q' \implies \mathbb{E}_Q\{\mu_{a,j,z}(V_j, U)\} = a + z = \mathbb{E}_{Q'}\{\mu_{a,j,z}(V_j, U)\}.$$

This proves all PROBE conditions for the designated orientation.

The outcome bridge $h(z^{\text{out}}, a) = a + z^{\text{out}}$ exists because

$$\mathbb{E}(Y \mid Z^{\text{tr}} = z, A = a) = a + z = \mathbb{E}\{h(Z^{\text{out}}, a) \mid Z^{\text{tr}} = z, A = a\}.$$

Canonical treatment-proxy completeness nevertheless fails. In the treated arm,

$$P(U_2 = 1 \mid U_1 = 0, A = 1) = \frac{11}{14}, \quad P(U_2 = 1 \mid U_1 = 1, A = 1) = \frac{15}{22}.$$

Define the square-integrable, nonzero function

$$g_1(U) = U_2 - \begin{cases} 11/14, & U_1 = 0, \\ 15/22, & U_1 = 1. \end{cases}$$

Then

$$\mathbb{E}\{g_1(U) \mid Z^{\text{tr}}, A = 1\} = \mathbb{E}\{g_1(U) \mid U_1, A = 1\} = 0$$

almost surely, although $g_1(U)$ is not almost surely zero. Hence the canonical completeness condition fails, proving the second direction. $\square$

Exhaustive plurality identification.

Proof of Theorem 2. Under condition (23), the number of proxy splits in $\mathcal{B}$ whose candidate value equals τ is strictly larger than the count for every non-target value. Therefore, τ is the unique value maximizing the population count:

$$\{\tau\} = \arg \max_{c \in \mathbb{R}} |\{S \in \mathcal{B} : \theta_S = c\}|.$$

If no non-target value occurs, the convention in Theorem 2 compares the nonempty target cluster with zero and gives the same conclusion. $\square$

Target-valued majority.

Proof of Corollary 1. Define

$$\begin{aligned} N_\tau &\triangleq |\{S \in \mathcal{B} : \theta_S = \tau\}|, \\ N_c &\triangleq |\{S \in \mathcal{B} : \theta_S = c\}|, \quad c \neq \tau. \end{aligned}$$

Condition (25) gives $N_\tau > |\mathcal{B}|/2$. For every $c \neq \tau$,

$$N_c \leq |\mathcal{B}| - N_\tau < N_\tau.$$

Therefore, τ is the unique candidate value of greatest multiplicity.

It remains to establish the median statement. Let $n = |\mathcal{B}|$. If $n = 2k + 1$ is odd, then $N_\tau \geq k + 1$. At most k candidate values are strictly below τ , and at most k are strictly above τ , so the $(k + 1)$ st order statistic equals τ . If $n = 2k$ is even, then $N_\tau \geq k + 1$ and $n - N_\tau \leq k - 1$. Hence both the k th and the $(k + 1)$ st order statistics equal τ . Under the conventional definition of the even-sample median as the average of these two middle order statistics, the median equals τ . Thus,

$$\text{median}((\theta_S)_{S \in \mathcal{B}}) = \tau.$$

The stronger condition that more than half of the proxy splits in $\mathcal{B}$ satisfy the grouped assumptions is sufficient because each such proxy split has candidate value τ , but it is not necessary. $\square$

Zero discrepancy and exact balance.

Proof of Proposition 2. By Definition 2, $D_{\text{res},j}(\phi) = 0$ if and only if

$$q_{j,\phi}(T_j, Z_j^\phi) = e_{j,\phi}(Z_j^\phi) \quad \text{almost surely.} \quad (75)$$

Because A is binary, this equality is equivalent to $T_j \perp\!\!\!\perp A \mid Z_j^\phi$. $\square$

Supporting bias decomposition. Fix a representation ϕ and abbreviate $e(z) = e_{j,\phi}(z)$. For each $a \in \{0, 1\}$ and $P_{Z_j^\phi}$ -almost every z , define

$$\delta_{a,j}^\phi(z) \triangleq \mathbb{E}_{Q_{1,j,z}^\phi} \{\mu_{a,j,z}^\phi(V_j, U)\} - \mathbb{E}_{Q_{0,j,z}^\phi} \{\mu_{a,j,z}^\phi(V_j, U)\}. \quad (76)$$

Under Assumptions 1 and 2, and under $0 < e(Z_j^\phi) < 1$ almost surely, the conditioning argument used in equation (68) gives

$$\mathbb{E}(Y \mid A = a, Z_j^\phi = z) = \mathbb{E}_{Q_{a,j,z}^\phi} \{\mu_{a,j,z}^\phi(V_j, U)\}. \quad (77)$$

Conditional expectation over A gives

$$\begin{aligned} \mathbb{E}\{Y(a) \mid Z_j^\phi = z\} &= e(z) \mathbb{E}_{Q_{1,j,z}^\phi} \{\mu_{a,j,z}^\phi\} \\ &\quad + \{1 - e(z)\} \mathbb{E}_{Q_{0,j,z}^\phi} \{\mu_{a,j,z}^\phi\}. \end{aligned} \quad (78)$$

Subtracting the two conditional expressions and collecting terms gives

$$\begin{aligned} &\mathbb{E}(Y \mid A = 1, Z_j^\phi = z) - \mathbb{E}(Y \mid A = 0, Z_j^\phi = z) \\ &\quad - [\mathbb{E}\{Y(1) \mid Z_j^\phi = z\} - \mathbb{E}\{Y(0) \mid Z_j^\phi = z\}] \\ &\quad = \{1 - e(z)\} \delta_{1,j}^\phi(z) + e(z) \delta_{0,j}^\phi(z). \end{aligned} \quad (79)$$

Approximate PROBE bias bound.

Proof of Theorem 3. Fix z in the almost-sure set on which overlap and the structural conditions hold. Bayes' rule gives

$$\frac{d\mathcal{L}(T_j \mid A = 1, Z_j^\phi = z)}{dP_{j,\phi,z}}(t) = \frac{q_{j,\phi}(t, z)}{e(z)}, \quad (80)$$

$$\frac{d\mathcal{L}(T_j \mid A = 0, Z_j^\phi = z)}{dP_{j,\phi,z}}(t) = \frac{1 - q_{j,\phi}(t, z)}{1 - e(z)}. \quad (81)$$

Subtracting gives the density identity

$$\frac{d\{\mathcal{L}(T_j \mid A = 1, Z_j^\phi = z) - \mathcal{L}(T_j \mid A = 0, Z_j^\phi = z)\}}{dP_{j,\phi,z}}(t) = \frac{q_{j,\phi}(t, z) - e(z)}{e(z)\{1 - e(z)\}}. \quad (82)$$

The common held-out channel gives $\mathcal{K}_j Q_{b,j,z}^\phi = \mathcal{L}(T_j \mid A = b, Z_j^\phi = z)$ for $b \in \{0, 1\}$. Assumption 4 therefore implies

$$|\delta_{a,j}^\phi(z)| \leq \frac{\Gamma_j(\phi)}{e(z)\{1 - e(z)\}} \left[\mathbb{E} \left(\{q_{j,\phi}(T_j, z) - e(z)\}^2 \mid Z_j^\phi = z \right) \right]^{1/2}. \quad (83)$$

Apply the triangle inequality to equation (79), use equation (83) for both potential outcomes, and integrate over Z_j^ϕ . Cauchy–Schwarz gives

$$\begin{aligned} & \left| \mathbb{E} \left[\mathbb{E}(Y \mid A = 1, Z_j^\phi) - \mathbb{E}(Y \mid A = 0, Z_j^\phi) \right] - \tau \right| \\ & \leq \mathbb{E} \left[\frac{\Gamma_j(\phi)}{e_{j,\phi}(Z_j^\phi)\{1 - e_{j,\phi}(Z_j^\phi)\}} \left\{ \mathbb{E} \left[\{q_{j,\phi}(T_j, Z_j^\phi) - e_{j,\phi}(Z_j^\phi)\}^2 \mid Z_j^\phi \right] \right\}^{1/2} \right] \end{aligned} \quad (84)$$

$$\leq \frac{\Gamma_j(\phi)}{\eta(1 - \eta)} D_{\text{res},j}(\phi). \quad (85)$$

Because $\Gamma_j(\phi)$ does not depend on z , the overlap bound $\eta \leq e_{j,\phi}(z) \leq 1 - \eta$ gives the last inequality and proves equation (34). $\square$

B.2. Proofs for Section 4

Uniform discrepancy deviation.

Proof of Proposition 4. For clarity, the joint encoder–critic loss classes used in the proposition are

$$\mathcal{L}_{0,j,n_D} \triangleq \left\{ o \mapsto \{a - h(\phi(v_j))\}^2 : \phi \in \mathcal{F}_{j,n_D}, h \in \mathcal{H}_{0,n_D} \right\}, \quad (86)$$

$$\mathcal{L}_{1,j,n_D} \triangleq \left\{ o \mapsto \{a - h(t_j, \phi(v_j))\}^2 : \phi \in \mathcal{F}_{j,n_D}, h \in \mathcal{H}_{1,n_D} \right\}, \quad (87)$$

where $o = (v_j, t_j, a, y)$. For a loss class $\mathcal{L}$, its expected Rademacher complexity is

$$\mathfrak{R}_{n_D}(\mathcal{L}) \triangleq \mathbb{E} \left[\sup_{\ell \in \mathcal{L}} \frac{1}{n_D} \sum_{i=1}^{n_D} \sigma_i \ell(O_i) \right], \quad (88)$$

where the σ_i are independent Rademacher signs independent of the observations.

Assumption 6 (Measurability for representation learning). Assume the two loss classes are pointwise measurable and separable, so the displayed suprema are measurable. Assume also that the selected approximate encoder and critic minimizers are measurable functions of the training observations and algorithmic seed.

The same argument may instead be stated with outer probability and outer expectation.

Every loss in either class lies in $[0, 1]$. Standard symmetrization and bounded-difference concentration therefore imply, simultaneously for $b \in \{0, 1\}$, that

$$\sup_{\phi \in \mathcal{F}_{j,n_D}, h \in \mathcal{H}_{b,n_D}} |\hat{R}_{b,\phi}(h) - R_{b,\phi}(h)| \leq 2\mathfrak{R}_{n_D}(\mathcal{L}_{b,j,n_D}) + \sqrt{\frac{\log(4/\delta_D)}{2n_D}} \quad (89)$$

with probability at least $1 - \delta_D$. Let $R_{b,\phi}^* = \inf_{h \in \mathcal{H}_{b,n_D}} R_{b,\phi}(h)$ and define $\hat{R}_{b,\phi}^*$ analogously. Equation (89) controls $|\hat{R}_{b,\phi}^* - R_{b,\phi}^*|$ by the same right-hand side. Equation (41) also gives

$$0 \leq \hat{R}_{b,\phi}(\hat{h}_{b,\phi}) - \hat{R}_{b,\phi}^* \leq \varepsilon_{b,\phi}. \quad (90)$$

The conditional-mean property of squared loss yields the Brier identity

$$\begin{aligned} & \mathbb{E}[\{A - e_{j,\phi}(Z_j^\phi)\}^2] - \mathbb{E}[\{A - q_{j,\phi}(T_j, Z_j^\phi)\}^2] \\ &= \mathbb{E}[\{q_{j,\phi}(T_j, Z_j^\phi) - e_{j,\phi}(Z_j^\phi)\}^2] = D_{\text{res},j}^2(\phi). \end{aligned} \quad (91)$$

It follows from equation (46) that

$$\left| R_{0,\phi}^* - R_{1,\phi}^* - D_{\text{res},j}^2(\phi) \right| \leq a_{0,n_D}(\phi) + a_{1,n_D}(\phi) \leq a_{j,n_D}. \quad (92)$$

Combine equations (89)–(92). The positive-part map is one-Lipschitz and $D_{\text{res},j}^2(\phi) \geq 0$, so taking the supremum over ϕ gives equation (47). $\square$

Representation-learning oracle inequality.

Proof of Proposition 5. On the event in equation (47), set

$$\xi_{j,n_D} \triangleq r_{j,n_D}(\delta_D) + a_{j,n_D} + \varepsilon_{\text{crit},j}. \quad (93)$$

The deviation inequality at $\hat{\phi}_j$, Equation (48), and the deviation inequality at a near-minimizer of $D_{\text{res},j}^2$ over $\mathcal{F}_{j,n_D}$ give

$$D_{\text{res},j}^2(\hat{\phi}_j) \leq \tilde{D}_{j,n_D}^2(\hat{\phi}_j) + \xi_{j,n_D} \quad (94)$$

$$\leq \inf_{\phi \in \mathcal{F}_{j,n_D}} \tilde{D}_{j,n_D}^2(\phi) + \varepsilon_{\text{enc},j} + \xi_{j,n_D} \quad (95)$$

$$\leq \inf_{\phi \in \mathcal{F}_{j,n_D}} D_{\text{res},j}^2(\phi) + 2\xi_{j,n_D} + \varepsilon_{\text{enc},j}, \quad (96)$$

which proves equation (49). $\square$

Causal error of the learned representation.

Proof of Theorem 4. Conditional on the training observations $\{O_i : i \in \mathcal{I}_D\}$ and any algorithmic seed used in training, the selected encoder is fixed; the population expectations below use a fresh observation independent of these training inputs. Equation (34) gives

$$|\tau_{\hat{Z}_j} - \tau| \leq \frac{\Gamma_j(\hat{\phi}_j)}{\eta(1-\eta)} D_{\text{res},j}(\hat{\phi}_j). \quad (97)$$

Taking the square root of equation (49) and substituting it into this display proves equation (51). $\square$

Exact signed nuisance remainder.

Proof of Proposition 6. Fix $\widehat{Z}_j = z$. Taking the conditional expectation of the score and using equation (52) gives

$$\begin{aligned} \mathbb{E}\{\text{UIF}_j(O) \mid \widehat{Z}_j = z\} &= \widehat{m}_{1,j}(z) - \widehat{m}_{0,j}(z) \\ &\quad + \frac{e_j(z)\{m_{1,j}(z) - \widehat{m}_{1,j}(z)\}}{\widehat{e}_j(z)} \\ &\quad - \frac{\{1 - e_j(z)\}\{m_{0,j}(z) - \widehat{m}_{0,j}(z)\}}{1 - \widehat{e}_j(z)}. \end{aligned} \quad (98)$$

Subtract $m_{1,j}(z) - m_{0,j}(z)$ and collect the two nuisance errors. Averaging the resulting identity over $\widehat{Z}_j$ proves equation (57). $\square$

Conditional finite-sample AIPW error.

Proof of Theorem 5. Conditional on the two learning samples, the evaluation observations are independent and the score $\text{UIF}_j(O)$ has variance σ_j^2 , so Chebyshev's inequality bounds the difference between the evaluation average and its conditional expectation by $\sigma_j/\sqrt{n_E\delta_E}$, the first term in equation (59), with probability at least $1 - \delta_E$. Proposition 6, Cauchy–Schwarz, and the propensity clipping bound control the absolute conditional remainder by

$$\frac{q_{e,j}(q_{0,j} + q_{1,j})}{\eta}. \quad (99)$$

The triangle inequality proves equation (60). For equation (61), the triangle inequality gives $|\widehat{\tau}_j^{\text{AIPW}} - \tau| \leq |\widehat{\tau}_j^{\text{AIPW}} - \tau_{\widehat{Z}_j}| + |\tau_{\widehat{Z}_j} - \tau|$. Equation (60) bounds the first term on the same event, and equation (34) bounds the second term by $\Gamma_j(\widehat{\phi}_j)D_{\text{res},j}(\widehat{\phi}_j)/\{\eta(1 - \eta)\}$. $\square$

Consistency.

Proof of Corollary 2. Condition (C1) makes the evaluation term $\sigma_{j,n}/\sqrt{n_E\delta_{E,n}}$ of $\varepsilon_{\text{AIPW},j}$ vanish, condition (C2) makes its nuisance-product term vanish, and condition (C3) makes the stability-weighted residual-discrepancy term in equation (61) vanish. The failure probability $\delta_{E,n}$ of Theorem 5 also converges to zero, which proves convergence in probability. $\square$

Random orientation search.

Proof of Theorem 6. Fix m with $P(M = m) > 0$. Conditional on $(M = m, R = r)$, the retained orientations form a uniformly sampled r -element subset of $\mathcal{B}$. For a non-target value c , define the finite-population variable

$$H_c(S) \triangleq 1\{\theta_S = \tau\} - 1\{\theta_S = c\}, \quad S \in \mathcal{B}.$$

It takes values in $[-1, 1]$, and its population mean satisfies

$$\frac{1}{N} \sum_{S \in \mathcal{B}} H_c(S) = p_\tau - \frac{n(c)}{N} \geq p_\tau - p_\star = \Delta.$$

Let $\overline{H}_{c,r}$ be its average over the retained orientations. The event that value c ties or defeats the target value implies $\overline{H}_{c,r} \leq 0$. Hoeffding's inequality for sampling without replacement therefore gives

$$\begin{aligned} P(\overline{H}_{c,r} \leq 0 \mid M = m, R = r) &\leq \exp \left\{ -\frac{2r\Delta^2}{(1 - (-1))^2} \right\} \\ &= \exp \left\{ -\frac{r\Delta^2}{2} \right\}. \end{aligned}$$

When $R = r \geq r_0$, a union bound over the L non-target values yields

$$P(\text{failure} \mid M = m, R = r) \leq L \exp \left\{ -\frac{r_0\Delta^2}{2} \right\}.$$

Combining this bound with

$$P(R < r_0 \mid M = m) \leq \eta_R(m)$$

gives

$$P(\text{success} \mid M = m) \geq \left[1 - \eta_R(m) - L \exp \left\{ -\frac{r_0\Delta^2}{2} \right\} \right]_+.$$

Here a tie and the event $R = 0$ are counted as failures. If $L = 0$, every orientation in $\mathcal{B}$ has candidate value τ , so any retained candidate returns τ and the procedure succeeds whenever $R \geq 1$. $\square$

If M is random and independent of the observed data and learning outcomes, then conditional on every realized $(M, R) = (m, r)$ the retained set remains a uniform r -element subset of $\mathcal{B}$. Therefore, if $P(R \geq r_0) \geq 1 - \eta_R$, the same argument after averaging over M gives the unconditional bound in the main text with $\eta_R(m)$ replaced by η_R . When $L = 0$, success occurs whenever $R \geq 1$.

B.3. Finite-Strata MMD Specialization

This subsection records the earlier finite-strata analysis as an independent specialization. It is not used by the continuous-representation results in the main text.

Fix a finite library $\mathcal{F}_j^{\text{fs}} \subseteq \Phi_j$ with $N_j^{\text{fs}} \triangleq |\mathcal{F}_j^{\text{fs}}|$. Every $\phi \in \mathcal{F}_j^{\text{fs}}$ is deterministic and takes values in a common finite set $\mathcal{S}_j$ of size M_j . Write $Z_j^\phi = \phi(V_j)$, let $\mathcal{R}_j(\phi)$ be its positive-mass strata, and set $p_{j,z}(\phi) = P(Z_j^\phi = z)$. Assume, uniformly over the library and $z \in \mathcal{R}_j(\phi)$,

$$P(Z_j^\phi = z) \geq p_{\min} > 0, \quad \eta \leq P(A = 1 \mid Z_j^\phi = z) \leq 1 - \eta \quad (100)$$

for some $\eta \in (0, 1/2]$. Let κ_j be a characteristic kernel on (X, W_j) with reproducing kernel Hilbert

space $\mathcal{H}_j$ and

$$\sup_v \sqrt{\kappa_j(v, v)} \leq \rho_j < \infty. \quad (101)$$

For $z \in \mathcal{R}_j(\phi)$, define

$$\Delta_{j,z}^{\text{MMD}}(\phi) \triangleq \text{MMD}_{\kappa_j} \left\{ \mathcal{L}(X, W_j \mid A = 1, Z_j^\phi = z), \mathcal{L}(X, W_j \mid A = 0, Z_j^\phi = z) \right\}, \quad (102)$$

$$D_j^{\text{MMD}}(\phi) \triangleq \sum_{z \in \mathcal{R}_j(\phi)} p_{j,z}(\phi) \Delta_{j,z}^{\text{MMD}}(\phi). \quad (103)$$

For $i \in \mathcal{I}_D$, let $Z_{j,i}^\phi = \phi(V_{j,i})$ and define

$$N_{a,z}^\phi \triangleq \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\}, \quad N_z^\phi \triangleq N_{0,z}^\phi + N_{1,z}^\phi, \quad (104)$$

$$\hat{p}_{j,z}(\phi) \triangleq \frac{N_z^\phi}{n_D}, \quad \hat{m}_{a,j,z}(\phi) \triangleq \frac{1}{N_{a,z}^\phi} \sum_{i \in \mathcal{I}_D} 1\{A_i = a, Z_{j,i}^\phi = z\} \kappa_j\{(X_i, W_{j,i}), \cdot\}. \quad (105)$$

When both treatment-arm cells are nonempty, set

$$\hat{\Delta}_{j,z}^{\text{MMD}}(\phi) \triangleq \|\hat{m}_{1,j,z}(\phi) - \hat{m}_{0,j,z}(\phi)\|_{\mathcal{H}_j}, \quad (106)$$

$$\hat{D}_j^{\text{MMD}}(\phi) \triangleq \sum_{z: N_z^\phi > 0} \hat{p}_{j,z}(\phi) \hat{\Delta}_{j,z}^{\text{MMD}}(\phi). \quad (107)$$

If an occupied stratum contains only one treatment arm, set $\hat{D}_j^{\text{MMD}}(\phi) = +\infty$.

The MMD specialization uses its own structural stability condition. For every $\phi \in \mathcal{F}_j^{\text{fs}}$, every positive-mass stratum z , every $a \in \{0, 1\}$, and all $Q, Q' \in \mathcal{Q}_{j,z}$, assume

$$|\mathbb{E}_Q\{\mu_{a,j,z}\} - \mathbb{E}_{Q'}\{\mu_{a,j,z}\}| \leq \Gamma_j^{\text{MMD}}(\phi) \text{MMD}_{\kappa_j}(\mathcal{K}_j Q, \mathcal{K}_j Q'), \quad \sup_{\phi \in \mathcal{F}_j^{\text{fs}}} \Gamma_j^{\text{MMD}}(\phi) \leq \bar{\Gamma}_j^{\text{MMD}} < \infty. \quad (108)$$

Proposition 7 (Finite-strata MMD learning bound). *Let $\delta_D \in (0, 1)$ and suppose*

$$n_D \geq \frac{8}{p_{\min} \eta} \log \left(\frac{8M_j N_j^{\text{fs}}}{\delta_D} \right). \quad (109)$$

Define

$$e_{j,n_D}^{\text{MMD}}(\delta_D) \triangleq \rho_j \left[2 + \sqrt{2 \log \left(\frac{4M_j N_j^{\text{fs}}}{\delta_D} \right)} \right] \sqrt{\frac{2}{n_D p_{\min} \eta}}, \quad (110)$$

$$r_{j,n_D}^{\text{MMD}}(\delta_D) \triangleq 2e_{j,n_D}^{\text{MMD}}(\delta_D) + 2\rho_j \sqrt{\frac{2 \left[M_j \log 2 + \log\{4N_j^{\text{fs}}/\delta_D\} \right]}{n_D}}. \quad (111)$$

Then, with probability at least $1 - \delta_D$,

$$\sup_{\phi \in \mathcal{F}_j^{\text{fs}}} \left| \widehat{D}_j^{\text{MMD}}(\phi) - D_j^{\text{MMD}}(\phi) \right| \leq r_{j,n_D}^{\text{MMD}}(\delta_D). \quad (112)$$

If $\widehat{\phi}_j^{\text{fs}}$ obeys

$$\widehat{D}_j^{\text{MMD}}(\widehat{\phi}_j^{\text{fs}}) \leq \inf_{\phi \in \mathcal{F}_j^{\text{fs}}} \widehat{D}_j^{\text{MMD}}(\phi) + \varepsilon_{\text{opt},j}^{\text{fs}}, \quad (113)$$

then, on the same event,

$$D_j^{\text{MMD}}(\widehat{\phi}_j^{\text{fs}}) \leq \inf_{\phi \in \mathcal{F}_j^{\text{fs}}} D_j^{\text{MMD}}(\phi) + 2r_{j,n_D}^{\text{MMD}}(\delta_D) + \varepsilon_{\text{opt},j}^{\text{fs}}, \quad (114)$$

$$|\tau_{\widehat{Z}_j^{\text{fs}}} - \tau| \leq \bar{\Gamma}_j^{\text{MMD}} \left[\inf_{\phi \in \mathcal{F}_j^{\text{fs}}} D_j^{\text{MMD}}(\phi) + 2r_{j,n_D}^{\text{MMD}}(\delta_D) + \varepsilon_{\text{opt},j}^{\text{fs}} \right], \quad (115)$$

where $\widehat{Z}_j^{\text{fs}} = \widehat{\phi}_j^{\text{fs}}(V_j)$, provided the primitive structural conditions and overlap used in the exact bias decomposition hold for every candidate.

Proof. Equation (109) and a multiplicative Chernoff bound imply that every candidate treatment-by-stratum cell contains at least $n_D p_{\min} \eta / 2$ observations except on an event of probability at most $\delta_D / 4$. Conditional on the cell assignments, bounded Hilbert-space mean concentration and a union bound over at most $2M_j N_j^{\text{fs}}$ cells give

$$\left| \widehat{\Delta}_{j,z}^{\text{MMD}}(\phi) - \Delta_{j,z}^{\text{MMD}}(\phi) \right| \leq 2e_{j,n_D}^{\text{MMD}}(\delta_D) \quad (116)$$

simultaneously, with failure probability at most $\delta_D / 2$. A multinomial L_1 concentration bound gives

$$\sum_z |\widehat{p}_{j,z}(\phi) - p_{j,z}(\phi)| \leq \sqrt{\frac{2 \left[M_j \log 2 + \log\{4N_j^{\text{fs}}/\delta_D\} \right]}{n_D}} \quad (117)$$

simultaneously over candidates, with failure probability at most $\delta_D / 4$. On the intersection of these events, $0 \leq \Delta_{j,z}^{\text{MMD}}(\phi) \leq 2\rho_j$ and the triangle inequality give equation (112).

The empirical-risk comparison yields

$$D_j^{\text{MMD}}(\widehat{\phi}_j^{\text{fs}}) \leq \widehat{D}_j^{\text{MMD}}(\widehat{\phi}_j^{\text{fs}}) + r_{j,n_D}^{\text{MMD}}(\delta_D) \quad (118)$$

$$\leq \inf_{\phi \in \mathcal{F}_j^{\text{fs}}} D_j^{\text{MMD}}(\phi) + 2r_{j,n_D}^{\text{MMD}}(\delta_D) + \varepsilon_{\text{opt},j}^{\text{fs}}, \quad (119)$$

which proves equation (114). The exact bias decomposition and the MMD stability condition in

equation (108) give

$$|\tau_{Z_j^\phi} - \tau| \leq \Gamma_j^{\text{MMD}}(\phi) D_j^{\text{MMD}}(\phi). \quad (120)$$

Apply this inequality to $\widehat{\phi}_j^{\text{fs}}$ and substitute equation (114) to obtain equation (115). $\square$